UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 9

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Theorem 1. *Given* $\mathbf{A} \in \mathbb{R}^{m\times n}$*, we have*

$$\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A}) = \sqrt{\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})},$$

where $\sigma_1(\mathbf{A})$ *is the maximum singular value of* $\mathbf{A}$ *and* $\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ *is the maximum eigenvalue of* $\mathbf{A}^{\mathrm{T}}\mathbf{A}$.

Interpretation: the maximum singular value acts as a norm.

Proof. Let λ_1 denote the maximum eigenvalue of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$, and $\boldsymbol{v}^{(1)}$ be its corresponding unit eigenvector. By the property of L_p-induced norm at the end of the previous lecture note,

$$\|\mathbf{A}\|_2 = \max_{\|\boldsymbol{w}\|_2=1} \|\mathbf{A}\boldsymbol{w}\|_2 \geq \|\mathbf{A}\boldsymbol{v}^{(1)}\|_2.$$

(Because optimal $\boldsymbol{w}$ is better than picking an arbitrary value $\boldsymbol{v}^{(1)}$.) Therefore,

$$\begin{aligned}\|\mathbf{A}\|_2 \geq \|\mathbf{A}\boldsymbol{v}^{(1)}\|_2 &= \sqrt{\langle \mathbf{A}\boldsymbol{v}^{(1)}, \mathbf{A}\boldsymbol{v}^{(2)}\rangle} \\ &= \sqrt{(\boldsymbol{v}^{(1)})^{\mathrm{T}} \underbrace{\mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{v}^{(1)}}_{\lambda_1 \boldsymbol{v}^{(1)}}} = \sqrt{\lambda_1 \underbrace{\|\boldsymbol{v}^{(1)}\|^2}_{=1}} = \sqrt{\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})}.\end{aligned}$$

It remains to show that $\|\mathbf{A}\|_2 \leq \sqrt{\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})}$. Let $\mathbf{A}^{\mathrm{T}}\mathbf{A} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{\mathrm{T}}$ be the eigenvalue decomposition of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$. Notice that

$$\begin{aligned}\|\mathbf{A}\|_2^2 &= \max_{\|\boldsymbol{w}\|_2=1} \|\mathbf{A}\boldsymbol{w}\|^2 = \max_{\|\boldsymbol{w}\|_2=1} \langle \mathbf{A}\boldsymbol{w}, \mathbf{A}\boldsymbol{w}\rangle \\ &= \max_{\|\boldsymbol{w}\|_2=1} \boldsymbol{w}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{w} = \max_{\|\boldsymbol{w}\|_2=1} \boldsymbol{w}^{\mathrm{T}}\mathbf{V}\boldsymbol{\Lambda} \underbrace{\mathbf{V}^{\mathrm{T}}\boldsymbol{w}}_{\text{call it } \boldsymbol{y}} \\ &= \max_{\|\boldsymbol{y}\|_2=1} \boldsymbol{y}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{y}.\end{aligned}$$

(Note: $\|\boldsymbol{y}\|_2^2 = \langle \boldsymbol{y}, \boldsymbol{y}\rangle = \boldsymbol{y}^{\mathrm{T}}\boldsymbol{y} = \boldsymbol{w}^{\mathrm{T}} \underbrace{\mathbf{V}\mathbf{V}^{\mathrm{T}}}_{\mathbf{I}} \boldsymbol{w} = \|\boldsymbol{w}\|_2^2 = 1$.)

Therefore,

$$\begin{aligned}\|\mathbf{A}\|_2^2 &= \max_{\|\boldsymbol{y}\|_2=1} [y_1, \ldots, y_n] \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \\ &= \max_{\|\boldsymbol{y}\|_2=1} \left(\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2\right) \\ &\leq \max_{\|\boldsymbol{y}\|_2=1} \lambda_1 \underbrace{(y_1^2 + \cdots + y_n^2)}_{\|\boldsymbol{y}\|_2^2} = \underbrace{\lambda_1}_{\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})}.\end{aligned}$$

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Let's get back to our original question:

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Given a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ (such as the grayscale image), approximate it as $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$ where $\boldsymbol{x} \in \mathbb{R}^m$, $\boldsymbol{y} \in \mathbb{R}^n$.

- Form an optimization:

$$\underbrace{\min_{\substack{\boldsymbol{x}\in\mathbb{R}^m\\ \boldsymbol{y}\in\mathbb{R}^n}} \|\mathbf{A}-\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}\|_{\mathrm{F}}}_{\text{optimization 1}} \quad \text{or} \quad \underbrace{\min_{\substack{\boldsymbol{x}\in\mathbb{R}^m\\ \boldsymbol{y}\in\mathbb{R}^n}} \|\mathbf{A}-\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}\|_{2}}_{\text{optimization 2}}.$$

- It happens that optimization 1 and 2 have very different objectives but give the same solution.
- First note that $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}} = (\alpha\boldsymbol{x})(\boldsymbol{y}/\alpha)^{\mathrm{T}}$ for all $\alpha \in \mathbb{R}$ and $\alpha \neq 0$.
- So, the optimal $(\boldsymbol{x}, \boldsymbol{y})$ is not unique due to scaling. But we hope that the optimal $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$ is unique.

Theorem 2. *A non-zero matrix $\mathbf{B} \in \mathbb{R}^{n\times m}$ has rank 1 iff there exist $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{y} \in \mathbb{R}^m$ such that $\mathbf{B} = \boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$. (If $\mathbf{B} = \mathbf{0}$, $\mathrm{rank}(\mathbf{B}) = 0$.)*

Proof. We prove one direction of Theorem 2.

- Assume $\mathbf{B}$ has rank 1. Let the columns of $\mathbf{B}$ be denoted as $\boldsymbol{b}^{(1)}, \ldots, \boldsymbol{b}^{(m)}$.
- $\mathrm{span}(\boldsymbol{b}^{(1)}, \ldots, \boldsymbol{b}^{(m)})$ has dimension 1.
- Let $\boldsymbol{x} \in \mathbb{R}^n$ be a basis for this one-dimensional space. Then, each column of $\mathbf{B}$ can be written as a linear combination of this basis:

$$\exists y_1, \ldots, y_m \in \mathbb{R} \text{ s.t. } \begin{cases} \boldsymbol{b}^{(1)} = y_1\boldsymbol{x}, \\ \quad\vdots \\ \boldsymbol{b}^{(m)} = y_m\boldsymbol{x} \end{cases}$$
$$\Rightarrow \mathbf{B} = \begin{bmatrix} \boldsymbol{b}^{(1)} & \cdots & \boldsymbol{b}^{(m)} \end{bmatrix} = \boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}.$$

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The previous question can be written as

$$\min_{\mathbf{B}\in\mathbb{R}^{m\times n}} \|\mathbf{A}-\mathbf{B}\|_{2 \text{ or } \mathrm{F}} \text{ subject to } \mathrm{rank}(\mathbf{B}) = 1. \tag{1}$$

This is the same as

$$\min_{\mathbf{B}\in\mathbb{R}^{m\times n}} \|\mathbf{A}-\mathbf{B}\|_{2 \text{ or } \mathrm{F}} \text{ subject to } \mathrm{rank}(\mathbf{B}) \leq 1. \tag{2}$$

(This is because $\mathbf{B} = \mathbf{0}$ can't be the optimal point if $\mathbf{A} \neq \mathbf{0}$; you can prove this using SVD.) Problem (2) is better than Problem (1) since if $\mathbf{A} = \mathbf{0}$, Problem (1) is not well-defined:

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Assume $m = n = 1$, $A = 0$. Problem (1) is

$$\begin{aligned} &\min_{B\in\mathbb{R}} |B| \ \text{ s.t. } \operatorname{rank}(B) = 1 \\ \Leftrightarrow &\min_{B\in\mathbb{R}} |B| \ \text{ s.t. } B \neq 0. \end{aligned}$$

The optimal objective value is 0, but there is no solution corresponding to that:

$$B = \varepsilon \quad \text{and} \quad \varepsilon \to 0.$$

To understand the applications of (2), consider the Netflix problem:

- Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ where m is the number of movies and n is the number of people.
- Entry a_{ij} is a number from 0 to 1 showing how much person j likes movie i.
- Example: if $m = 100,000$, $n = 500,000,000$ (half a billion), then the data in $\mathbf{A}$ is $50,000\,\text{B}$.
- If $\mathbf{A}$ is approximated as $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$, then the data size is (size of $\boldsymbol{x}$) + (size of $\boldsymbol{y}$) $= m + n <$ 1 B.
- $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$ means all columns of the matrix differ by some scaling factor:

$$\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}} = \begin{bmatrix} \boldsymbol{x}y_1 & \boldsymbol{x}y_2 & \cdots & \boldsymbol{x}y_n \end{bmatrix}.$$

- That means the movie preferences of all people should be aligned through some universal vector of preferences $\boldsymbol{x}$, which is not realistic.

What if we revise (2) as

$$\min_{\mathbf{B}\in\mathbb{R}^{m\times n}} \|\mathbf{A} - \mathbf{B}\|_{2 \text{ or } \mathrm{F}} \ \text{ subject to } \ \operatorname{rank}(\mathbf{B}) \leq k,$$

for a given k that is close to 1, rather than $\min(m, n)$?
(Assume $k \leq \min(m, n)$. Since $\mathbf{B} \in \mathbb{R}^{m\times n}$, $\operatorname{rank}(\mathbf{B})$ can't be more than $\min(m, n)$.)
This problem is called *low-rank matrix approximation.*
First, let's try to understand the condition $\operatorname{rank}(\mathbf{B}) \leq k$.

Theorem 3. *(i) Matrix $\mathbf{B} \in \mathbb{R}^{m\times n}$ satisfies $\operatorname{rank}(\mathbf{B}) \leq k$ iff there are matrices $\mathbf{C} \in \mathbb{R}^{m\times k}$ and $\mathbf{D} \in \mathbb{R}^{n\times k}$ such that $\mathbf{B} = \mathbf{C}\mathbf{D}^{\mathrm{T}}$.*

(ii) Matrix $\mathbf{B} \in \mathbb{R}^{m\times n}$ satisfies $\operatorname{rank}(\mathbf{B}) \leq k$ iff there are vectors $\boldsymbol{c}^{(1)}, \ldots, \boldsymbol{c}^{(k)} \in \mathbb{R}^m$ and $\boldsymbol{d}^{(1)}, \ldots, \boldsymbol{d}^{(k)} \in \mathbb{R}^n$ such that

$$\mathbf{B} = \underbrace{\boldsymbol{c}^{(1)}\left(\boldsymbol{d}^{(1)}\right)^{\mathrm{T}}}_{\text{rank } 1} + \underbrace{\boldsymbol{c}^{(2)}\left(\boldsymbol{d}^{(2)}\right)^{\mathrm{T}}}_{\text{rank } 1} + \cdots + \underbrace{\boldsymbol{c}^{(k)}\left(\boldsymbol{d}^{(k)}\right)^{\mathrm{T}}}_{\text{rank } 1}.$$

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Interpretation of part (ii): $\mathbf{B}$ has rank k or less iff it is the sum of k rank-1 matrices. Note: the sum of k rank-1 matrices may have rank $0, 1, \ldots, k$. Example:

$$\underbrace{\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix}}_{\text{rank 1}} + \underbrace{\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}-1 & -1\end{bmatrix}}_{\text{rank 1}} = \underbrace{\begin{bmatrix}0\\0\end{bmatrix}\begin{bmatrix}0 & 0\end{bmatrix}}_{\text{rank 0}}.$$

Proof. Part (ii) follows from part (i) since

$$\mathbf{C}\mathbf{D}^{\mathrm{T}} = \sum_{i=1}^{k} \boldsymbol{c}^{(i)} \left(\boldsymbol{d}^{(i)}\right)^{\mathrm{T}}.$$

- *Necessity proof of part (i).*
  Assume $\mathbf{B}$ has rank k or less. Use SVD:
  $$\mathbf{B} = \underbrace{\mathbf{U}}_{m\times m}\underbrace{\mathbf{\Sigma}}_{m\times n}\underbrace{\mathbf{V}^{\mathrm{T}}}_{n\times n},$$
  where $\mathbf{\Sigma}$ is a rectangular matrix with $\sigma_1, \ldots, \sigma_r, \underbrace{0, \ldots, 0}_{\min(m,n)-r}$ on the diagonal.
  Recall that $r = \mathrm{rank}(\mathbf{B}) \le k$. We focus on the top k singular values $\underbrace{\sigma_1, \ldots, \sigma_r, 0, \ldots, 0}_{k}$:
  $$\mathbf{B} = \underbrace{\begin{bmatrix}\boldsymbol{u}^{(1)} & \cdots & \boldsymbol{u}^{(k)}\end{bmatrix}\begin{bmatrix}\sigma_1 & & & & & \\ & \ddots & & & & \\ & & \sigma_r & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0\end{bmatrix}}_{\text{call this } \mathbf{C}}\underbrace{\begin{bmatrix}\left(\boldsymbol{v}^{(1)}\right)^{\mathrm{T}}\\ \vdots \\ \left(\boldsymbol{v}^{(k)}\right)^{\mathrm{T}}\end{bmatrix}}_{\text{call this } \mathbf{D}^{\mathrm{T}}}.$$
- *Sufficiency proof of part (i).*
  Assume $\mathbf{B} = \mathbf{C}\mathbf{D}^{\mathrm{T}}$ for some $\mathbf{C} \in \mathbb{R}^{m\times k}$ and $\mathbf{D} \in \mathbb{R}^{n\times k}$. Recall:
  $$\dim \mathcal{N}(\mathbf{B}) + \mathrm{rank}(\mathbf{B}) = n. \tag{3}$$
  Consider $\mathrm{span}(\boldsymbol{d}^{(1)}, \ldots, \boldsymbol{d}^{(k)})$. This is a subspace in $\mathbb{R}^n$ of dimension at most k. Thus,
  orthogonal space of this span subspace has dimension at least $n-k$
  $\Rightarrow \exists$linearly independent $\boldsymbol{w}^{(1)}, \ldots, \boldsymbol{w}^{(n-k)} \in \mathbb{R}^n$ s.t. $\langle \boldsymbol{d}^{(i)}, \boldsymbol{w}^{(j)} \rangle = 0,\ \forall i \le k, j \le n-k$
  $\Rightarrow \mathbf{D}^{\mathrm{T}}\boldsymbol{w}^{(j)} = \mathbf{0},\ \forall j \le n-k$
  $\Rightarrow \mathbf{B}\boldsymbol{w}^{(j)} = \mathbf{C}\mathbf{D}^{\mathrm{T}}\boldsymbol{w}^{(j)} = \mathbf{0},\ \forall j \le n-k$
  $\Rightarrow \dim \mathcal{N}(\mathbf{B}) \ge n-k.$ (4)
  Combining Equation (3) and Equation (4) gives
  $$\mathrm{rank}(\mathbf{B}) \le n - (n-k) = k.$$

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Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$ is approximated by a matrix $\mathbf{B}$ such that $\text{rank}(\mathbf{B}) \leq k$:

$$\mathbf{B} = \boldsymbol{c}^{(1)} \left(\boldsymbol{d}^{(1)}\right)^{\mathrm{T}} + \cdots + \boldsymbol{c}^{(k)} \left(\boldsymbol{d}^{(k)}\right)^{\mathrm{T}}.$$

- The amount of data in $\mathbf{B}$ is $k(m+n)$, while the amount of data in $\mathbf{A}$ is mn.
- If k is close to 1, this leads to a huge reduction in data.

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